

C3MAT1T1 – O – 25 – 4213



THIRD SEMESTER B.A./B.SC. (REVISED NEP) DEGREE
EXAMINATION, JANUARY 2026

MATHEMATICS

Algebra II and Real Analysis – I (DSC)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer **all** questions.

I. Answer **all** questions of the following :

(10×2=20)

- 1) Show that the identity element of a group is unique.
- 2) Define the left and right cosets of subgroup H of group G.
- 3) Define transitive relation and give an example.
- 4) Show that set of all integers is countable.
- 5) Define limit of a sequence.
- 6) Find $\lim_{n \rightarrow \infty} (n!)^{1/n}$.
- 7) State Cauchy's general principle of convergence.
- 8) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$.
- 9) State Cauchy's root test for series of positive terms.
- 10) Test the convergence of $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots$

II. Answer **any six** of the following :

(6×5=30)

- 11) State and prove Lagrange's theorem for a finite group.
- 12) Prove that a non empty subset H of a group G is a subgroup of G if and only if $ab^{-1} \in H$ whenever $a, b \in H$.
- 13) Prove that unit interval $[0, 1]$ is uncountable.
- 14) With usual notation, show that $N \times N$ is countable.

[P.T.O.]



15) If $\{a_n\}$ and $\{b_n\}$ are any two sequences, then prove that

$$\lim_{n \rightarrow \infty} (a_n b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n.$$

16) Prove that monotonic decreasing sequence which is bounded below converges to its infimum.

17) State and prove D'Alembert's ratio test for the convergence of series of positive terms.

18) Discuss the convergence of the series $\sum (\sqrt{n^2 + 1} - \sqrt{n^2 - 1})$.

III. Answer **any three** of the following :

(3×10=30)

19) a) Show that every subgroup of cyclic group is cyclic.

b) Show that the set Q_{-1} of all rational numbers except -1 with binary operation defined by $a*b = a+b+ab \forall a, b \in Q_{-1}$ is an abelian group.

20) a) Prove that the union of two countable set is countable.

b) State and prove Archimedian property of real numbers.

21) a) State and prove Cauchy's second theorem on limit of a sequence.

b) Prove that $\lim_{n \rightarrow \infty} \left(\frac{n^n}{n!} \right)^{\frac{1}{n}} = e$.

22) a) State and prove Leibnitz's test for alternating series.

b) Test the convergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots, x > 0$.
